Core Finance
Week 2

MGMP 543

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Spring 2025

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Welcome Back! Today's Topics

- Week 1 Recap
- Capital Budgeting Tools
- Bonds

Week 2 Introductions

Week 2 Introductions

- Name
- What’s the best thing you’ve done in the last week (month/since November)?

Week 1 Recap

What Did We Learn Last Week?

What Did We Learn Last Week?

- Time value of money ($1 today > $1 later)
- Value of assets is the stream of future value it will create in the future.
- Future Value (FV) and Present Value (PV) of assets
- Discounting and compound interest
- Perpetuities, Growing Perpetuities, Annuities, and Growing Annuities

You would like to have enough money saved to receive an $80,000 per year perpetuity after retirement. How much would you need to have saved in your retirement fund to achieve this goal? Assume that the perpetuity payments start on the day of your retirement. The annual interest rate is 10 percent.

Capital Budgeting Tools

How Does Your Firm Decide on Projects?

Capital Budgeting

Capital budgeting is process for evaluating major projects/investments.

- Evaluate potential investments
- Allocate funds over time towards preferred investments

Capital budget for decisions involving time, uncertainty, & cash flow measurable.

Financial managers should accept projects creating **value**

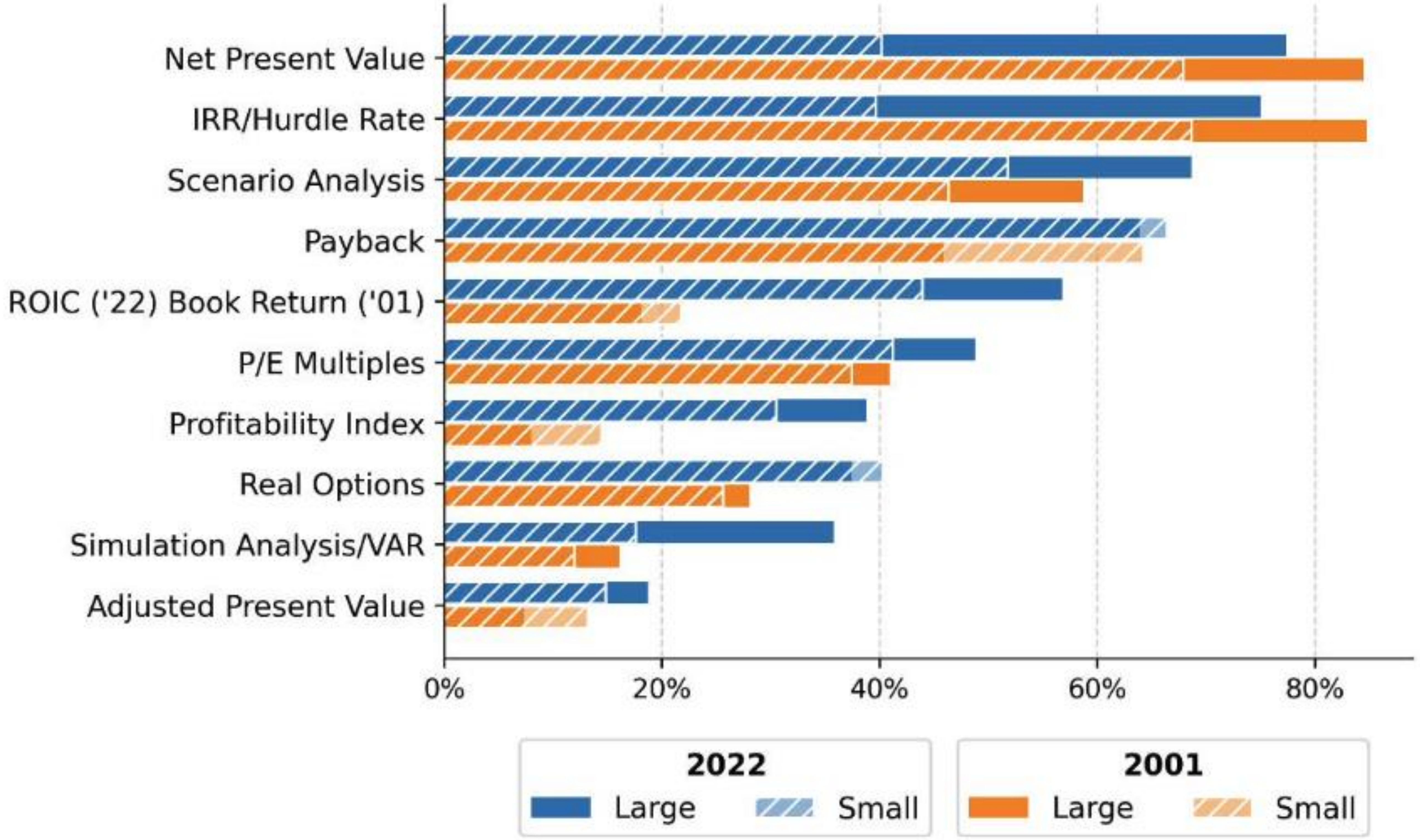
(expected benefits > expected costs)

Capital Budgeting Tools

1. Net Present Value (NPV)
2. Internal Rate of Return (IRR)
3. Payback Period (PP)
4. Discounted Payback Period (DPP)
5. Return on Investment (ROI)
6. Profitability Index (PI)
7. Equivalent Annual Cost (EAC)

Which Capital Budgeting Tools Are Actually Used?

Percent of CFO reporting always/almost always use a particular capital budgeting tool when deciding which projects/acquisitions to pursue



John Graham (2022). Large firms are $1bn+ revenue.

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Capital Budgeting Tools:
Net Present Value (NPV)

Capital Budgeting Tools

1. **Net Present Value (NPV)**
2. Internal Rate of Return (IRR)
3. Payback Period (PP)
4. Discounted Payback Period (DPP)
5. Return on Investment (ROI)
6. Profitability Index (PI)
7. Equivalent Annual Cost (EAC)

Does this look familiar?

$$\sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

Spot the Difference: From PV to NPV

Spot the Difference: From PV to NPV

$$NPV = \sum_{t=0}^{N} \frac{CF_t}{(1+r)^t}$$

$$NPV = -I_0 + \sum_{t=0}^{N} \frac{CF_t}{(1+r)^t}$$

PV, we are valuing cashflows in (or out).

NPV is more general version, valuing cashflows in and cashflows out.

Net Present Value (NPV)

$$NPV = \sum_{t=0}^{N} \frac{CF_t}{(1+r)^t}$$

Present value of all cash inflows minus present value of cash outflows

(i.e., discounted cash generated by investment, net of discounted costs of investment)

Decision Rule: Invest if and only if the project has a **positive** NPV

- If NPV > 0 : **Invest** ✅
- If NPV = 0: **Indifferent**
- If NPV < 0: **Don't Invest** ❌

NPV is the most important capital budgeting method

Capital Budgeting Tools: Internal Rate of Return (IRR)

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Spot the Difference: From NPV to IRR

$$NPV = \sum_{t=0}^{N} \frac{CF_t}{(1+r)^t}$$

Decision Rule: Invest if and only if

The Internal Rate of Return (IRR) is greater than the required return on the investment.

Spot the Difference: From NPV to IRR

$$NPV = \sum_{t=0}^{N} \frac{CF_t}{(1+r)^t}$$

$$0 = \sum_{t=0}^{N} \frac{CF_t}{(1+IRR)^t}$$

Decision Rule: Invest if and only if

The Internal Rate of Return (IRR) is greater than the required return on the investment.

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Internal Rate of Return (IRR) & Hurdle Rates

$$0 = \sum_{t=0}^{N} \frac{CF_t}{(1 + IRR)^t}$$

Decision Rule: Invest if and only if

The Internal Rate of Return (IRR) is greater than the required return on the investment.

Internal Rate of Return (IRR) & Hurdle Rates

$$0 = \sum_{t=0}^{N} \frac{CF_t}{(1 + IRR)^t}$$

Decision Rule: Invest if and only if

The Internal Rate of Return (IRR) is greater than the required return on the investment.

If IRR>r then invest. r is called the "Hurdle Rate"

(HR or Minimum Accepted Rate of Return, MARR)

A Few Reasons to be Wary of IRR

1. Scale
2. Time
3. Signs
4. Multiple IRRs
5. No IRR

IRR Issue 1: Scale

Two mutually exclusive projects A and B.

Project A has higher IRR but lower NPV than Project B.

Unlike NPV, IRR doesn't take different scales into account.

Project	CF_0	CF_1	NPV (r = 10%)	IRR
A	-10,000	12,000	909.09	
B	-80,000	92,000	3,363.36	

IRR Issue 1: Scale

Two mutually exclusive projects A and B.

Project A has higher IRR but lower NPV than Project B.

Unlike NPV, IRR doesn't take different scales into account.

Project	CF_0	CF_1	NPV (r = 10%)	IRR
A	-10,000	12,000	909.09	20%
B	-80,000	92,000	3,363.36	15%

IRR Issue 2: Time

Two mutually exclusive projects A and B.

Projects A and B have same size but different payoff timing.

Same IRR for both projects, but NPV varies depending on r.

Project	CF_0	CF_1	CF_2	IRR	NPV (r = 5%)	NPV (r = 10%)	NPV (r = 11%)
A	-10,000	1,000	11,000	10%	929.70	0.00	-171.25
B	-10,000	5,762	5,762	10%	3,363.36	0.17	-132.44

IRR Issue 3: Investing or Financing

Project A has investment-like cash flows (invest today, get return over time).

Project B has loan-like cash flows (borrow today, repay over time).

Projects A and B have same IRR but different NPVs.

Project	CF_0	CF_1	NPV (r = 10%)	IRR
A	-400	500	54.54	
B	400	-500	-54.54	

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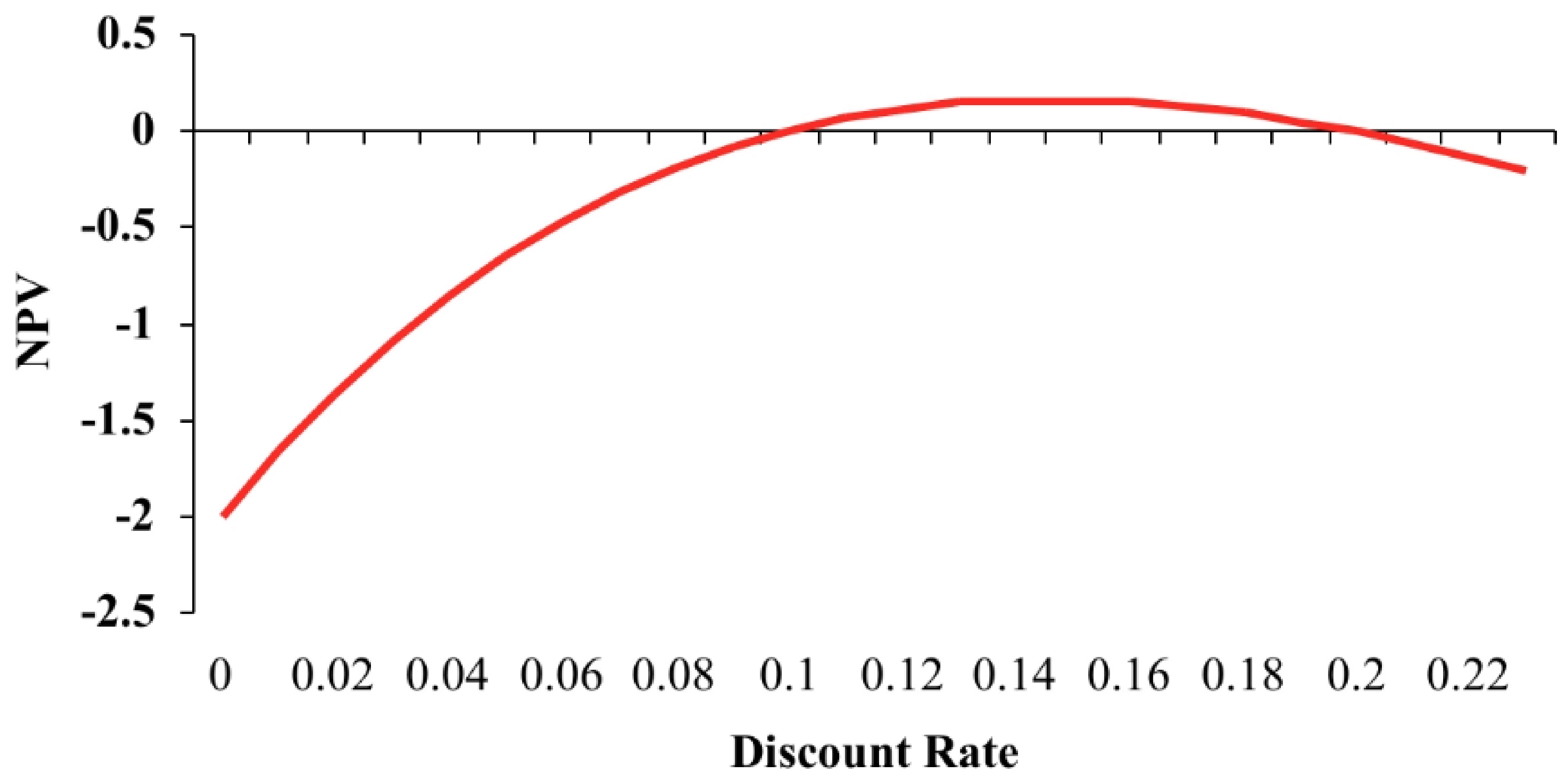
Project	CF_0	CF_1	NPV (r = 10%)	IRR
A	-400	500	54.54	25%
B	400	-500	-54.54	25%

IRR Issue 4: Multiple IRRs

In general, there will be as many IRRs as there are changes in signs of cash flows

Project	CF_0	CF_1	CF_2	NPV (r = 10%)	IRR
A	-100	230	-132	0	10% or 20%



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IRR Issue 5: No IRR

IRR undefined.

Project	CF_0	CF_1	CF_2	NPV (r = 10%)	IRR
A	-1,000	1,500	-1,000	-462.81	Undefined

When to use IRR?

Only if comparing opportunities with the same scale, timing, and risk

Capital Budgeting Tools: Payback Period (PP)

Payback Period (PP)

Minimum time ($\boldsymbol{T_{PP}}$) to recoup initial investment ($\boldsymbol{I_0}$)

Payback Period (PP)

Minimum time (T_{PP}) to recoup initial investment (I_0)

$$\sum_{t=1}^{T_{PP}} CF_t \geq I_0$$

Drawbacks?

- Ignores cash flows after payback period
- Ignores timing and risk of cashflows
- Arbitrary decision rule
- Doesn't distinguish projects with same payback period
- May suggest doing NPV < 0 projects!

Example where using Payback Period (PP) would recommend NPV<0 project

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV (r = 10%)	PP (years)
A	-100	109	0	0	0	-0.91	
B	-100	0	140	140	0	120.89	
C	-100	70	70	140	0	126.67	
D	-100	0	0	200	-220	-100.00	
E	-100	0	0	0	400	173.21	
F	-200	0	0	0	800	346.41	

Example where using Payback Period (PP) would recommend NPV<0 project

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV (r = 10%)	PP (years)
A	-100	109	0	0	0	-0.91	1
B	-100	0	140	140	0	120.89	2
C	-100	70	70	140	0	126.67	2
D	-100	0	0	200	-220	-100.00	3
E	-100	0	0	0	400	173.21	4
F	-200	0	0	0	800	346.41	4

Capital Budgeting Tools: Discounted Payback Period (DPP)

Discounted Payback Period (DPP)

Minimum time ($\boldsymbol{T_{DPP}}$) to recoup initial investment ($\boldsymbol{I_0}$) given discounted cash flows

Discounted Payback Period (DPP)

Minimum time ($\boldsymbol{T_{DPP}}$) to recoup initial investment ($\boldsymbol{I_0}$) given discounted cash flows

$$\sum_{t=1}^{T_{DPP}} \frac{CF_t}{(1+r)^t} \geq I_0$$

Drawbacks?

- Ignores cash flows after payback period
- ~~Ignores timing and risk of cashflows~~
- Arbitrary decision rule
- Doesn't distinguish projects with same payback period
- May suggest doing NPV < 0 projects

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Example where using Discounted Payback Period (DPP) wouldn't recommend highest NPV projects

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV (r = 10%)	DPP (years)
A	-100	109	0	0	0	-0.91	
B	-100	0	140	140	0	120.89	
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D	-100	0	0	200	-220	-100.00	
E	-100	0	0	0	400	173.21	
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Example where using Discounted Payback Period (DPP) wouldn't recommend highest NPV projects

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV (r = 10%)	DPP (years)
A	-100	109	0	0	0	-0.91	Infinite!
B	-100	0	140	140	0	120.89	2
C	-100	70	70	140	0	126.67	2
D	-100	0	0	200	-220	-100.00	3
E	-100	0	0	0	400	173.21	4
F	-200	0	0	0	800	346.41	4

Capital Budgeting Tools: Return on Investment (ROI)

Return on Investment (ROI)

$$\boldsymbol{ROI = \frac{Average\ Forecasted\ Profit}{Average\ Net\ Investment}}$$

Drawbacks?

- Ignores timing and risk of cashflows
- Arbitrary decision rule
- Depends on accounting income not on cash flows of project

Capital Budgeting Tools: Profitability Index (PI)

Profitability Index (PI)

A firm sometimes cannot do all NPV>0 projects, often due to resource constraints.

Profitability Index (PI) ranks projects by their “bang for your buck”

Firm should do all NPV>0 projects, moving down PI ranking, until all available resources are consumed.

$$\boldsymbol{PI = \frac{Value\ Created}{Resource\ Consumed} = \frac{NPV}{Resource\ Consumed}}$$

Challenges of Profitability Index (PI)

Two conditions required to use PI:

1. Need to exhaust available resources
2. Only one relevant dimension of resource constraints.

But if multiple resource constraints (e.g., budget and headcount)

PI also assumes that resources are fixed. But NPV>0 projects are worthwhile as they create value. Couldn't you expand resources to create value?

Capital Budgeting Tools: Equivalent Annual Cost (EAC)

Equivalent Annual Cost (EAC) makes NPV of mutually-exclusive projects with different lives more comparable

Equivalent Annual Cost (EAC) is an annuity with the same life and present value as the project

- Step 1. Calculate NPV
- Step 2. Divide NPV by the present value of an annuity factor (A) based on the project's discount rate (r) and duration (T)

Equivalent Annual Cost (EAC) makes NPV of mutually-exclusive projects with different lives more comparable

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- Step 1. Calculate NPV
- Step 2. Divide NPV by the present value of an annuity factor (A) based on the project's discount rate (r) and duration (T)

$$\boldsymbol{EAC = \frac{NPV}{A} \; where \; A = \frac{\left(1 - \frac{1}{(1+r)^T}\right)}{r}}$$

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Example of Equivalent Annual Cost (EAC)

$$EAC = \frac{NPV}{A} \ where \ A = \frac{\left(1 - \frac{1}{(1+r)^T}\right)}{r}$$

Project	CF_0	CF_1	CF_2	CF_3	NPV (r = 10%)	EAC
A	-100	110	170	0	90.91	
B	-100	110	121	133	199.92	

Example of Equivalent Annual Cost (EAC)

$$\boldsymbol{EAC = \frac{NPV}{A} \; where \; A = \frac{\left(1 - \frac{1}{(1+r)^T}\right)}{r}}$$

Project	CF_0	CF_1	CF_2	CF_3	NPV (r = 10%)	EAC
A	-100	110	170	0	90.91	52.38
B	-100	110	121	133	199.92	80.39

Bonds

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Bonds: Introductions

How Many Bonds Outstanding Worldwide?

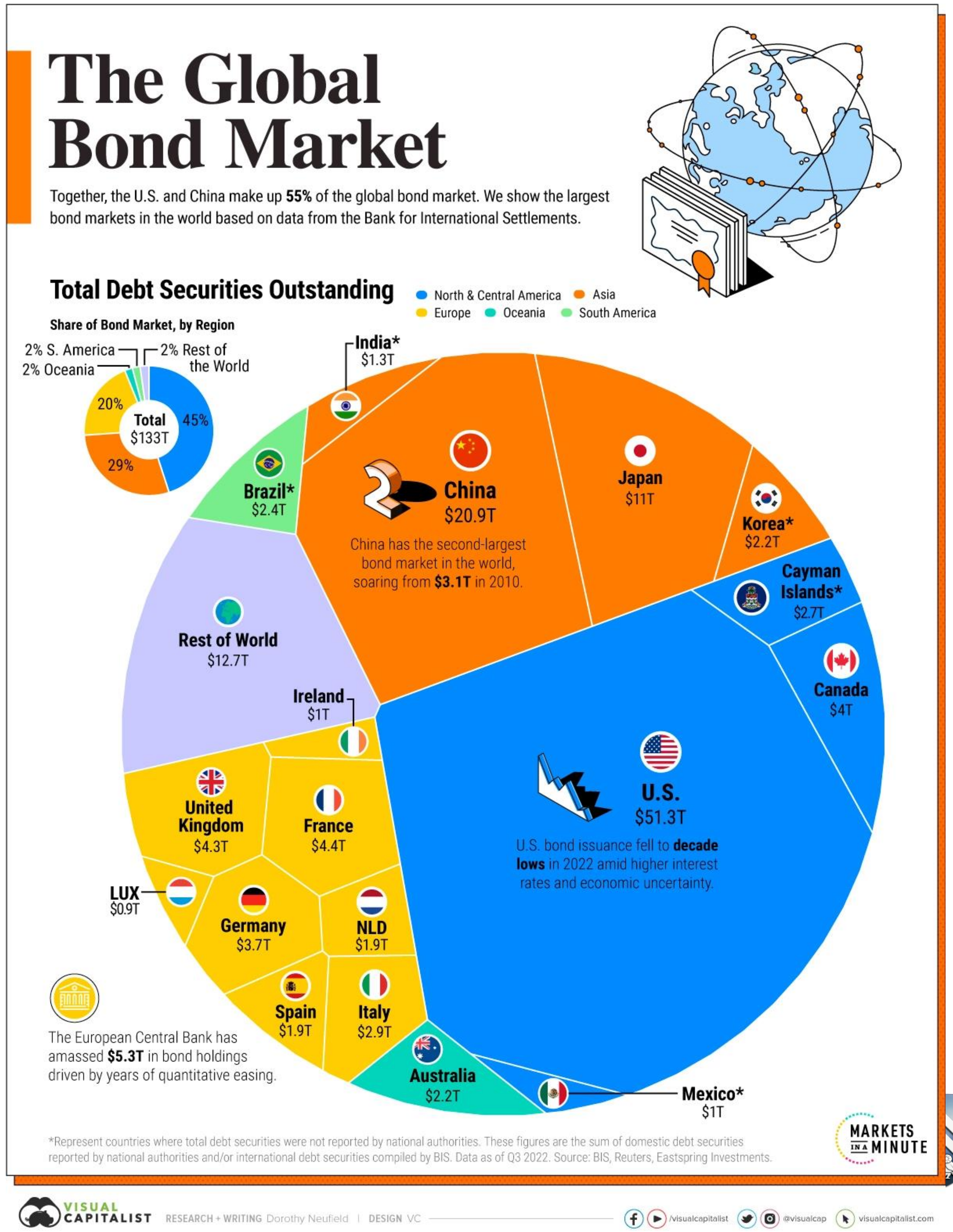
How Many Bonds Outstanding Worldwide? 4

RIP RIP

Next Bond?

How Much Bonds Outstanding Worldwide?

~$141 trillion in 2023 (SIFMA)



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Jargon-Busting Bonds

Example: Benedict issues an 8-year 9% coupon $1,000 face value bond, with semi-annual coupon payments.

A bond is a contract between an issuer and a bondholder.

Face Value (F): Amount to be repaid at maturity.
Also known as Principal.

Coupon Rate (CR): Stipulated interest rate to be paid on
the face value of a bond.

Coupon Payment (C): Periodic interest payment to bondholder.
6 months typical.

Time to maturity (n): Time until bond face value is due to be repaid.

Bond Price (B): Value of a bond today.

Covenants: Terms of the bond.

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Face Value (F): Amount to be repaid at maturity.
Also known as Principal. **$1,000.**

Coupon Rate (CR): Stipulated interest rate to be paid on
the face value of a bond. **9%.**

Coupon Payment (C): Periodic interest payment to bondholder.
6 months typical. **$45**

Time to maturity (n): Time until bond face value is due to be repaid. **8 years.**

Bond Price (B): Value of a bond today. **$???**

Covenants: Terms of the bond.

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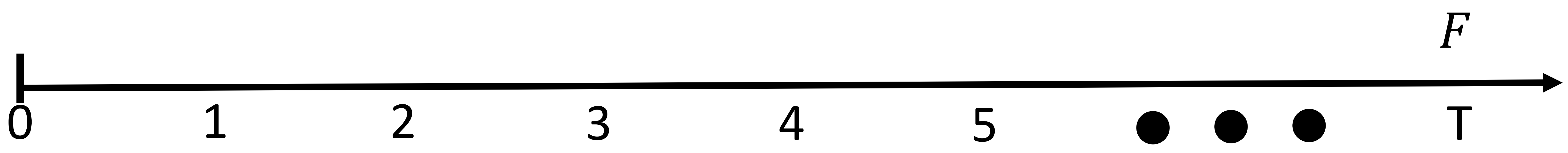
Bonds:
Pricing

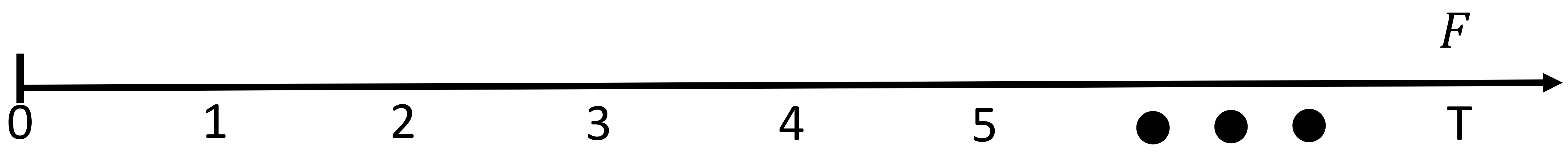
Valuing a Zero-Coupon Bond



What is the price of a zero-coupon bond which matures in 15 years and pays \$1,000 if the required rate of return is 9%?

Valuing a Zero-Coupon Bond

$$B = \frac{F}{(1+r)^T}$$



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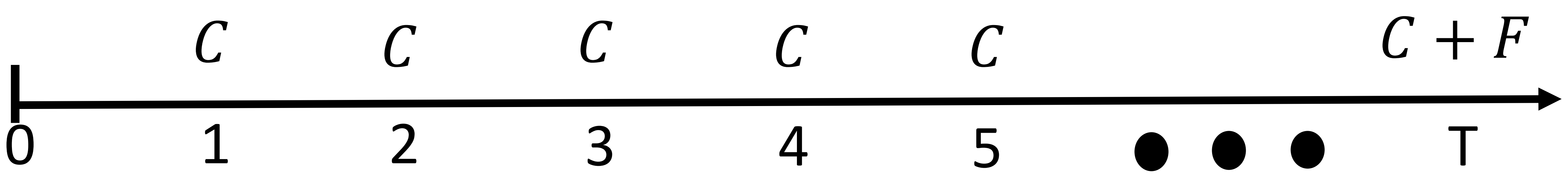
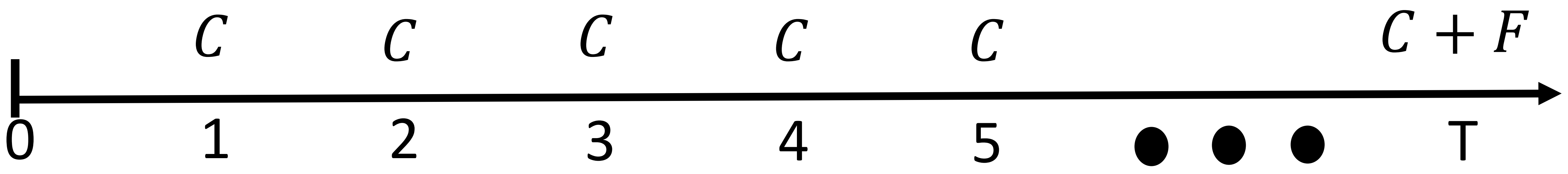
Valuing a Coupon Bond



What is the price of a 9% annual coupon bond with a face value of $1,000 which matures in 15 years if the required rate of return is 9%?

Valuing a Coupon Bond



$$B = \sum_{t=1}^{T} \frac{C}{(1+r)^t} + \frac{F}{(1+r)^T}$$

$$= \frac{C}{r}\left(1 - \frac{1}{(1+r)^T}\right) + \frac{F}{(1+r)^T}$$

What is the price of a 9% annual coupon bond with a face value of $1,000 which matures in 15 years if the required rate of return is 9%?

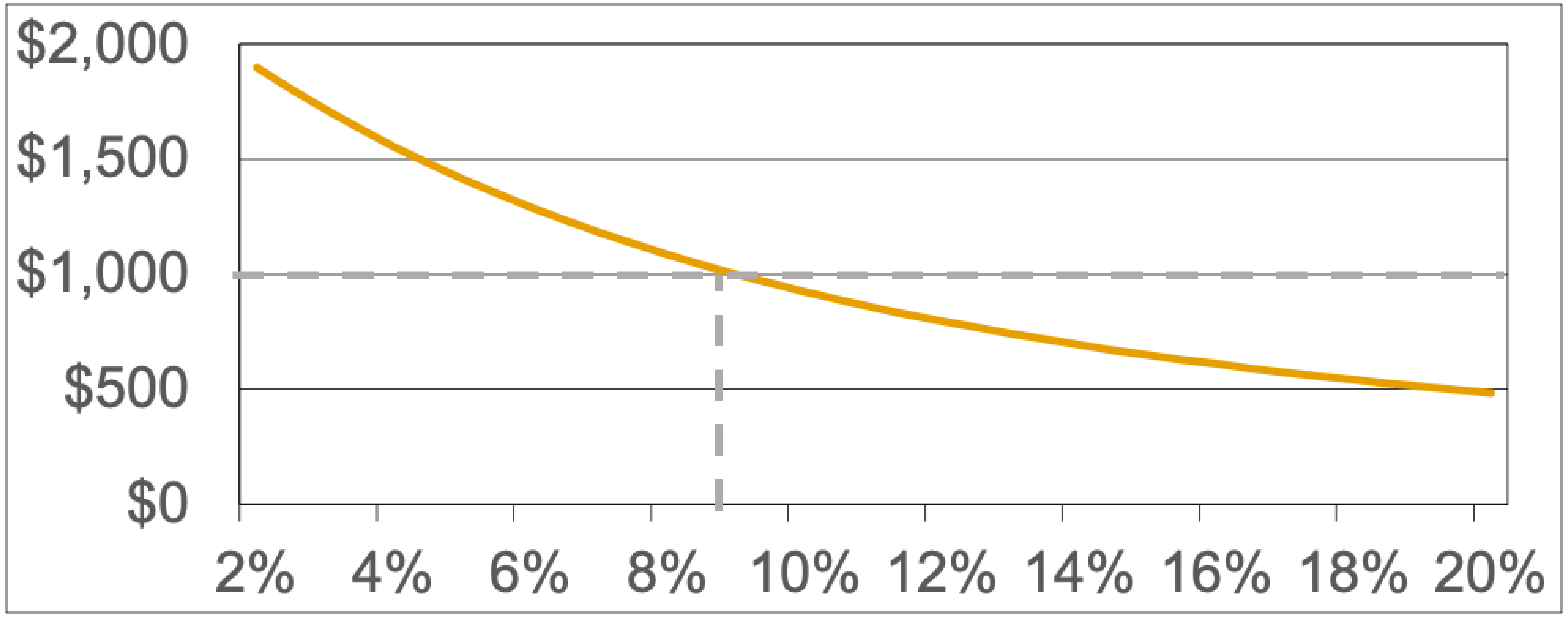
Bond prices when interest rates increase?

$$\boldsymbol{B = \frac{C}{r}\left(1 - \frac{1}{(1+r)^T}\right) + \frac{F}{(1+r)^T}}$$

Bond prices move ***inversely*** with interest rates

$$B = \frac{C}{r}\left(1 - \frac{1}{(1+r)^T}\right) + \frac{F}{(1+r)^T}$$

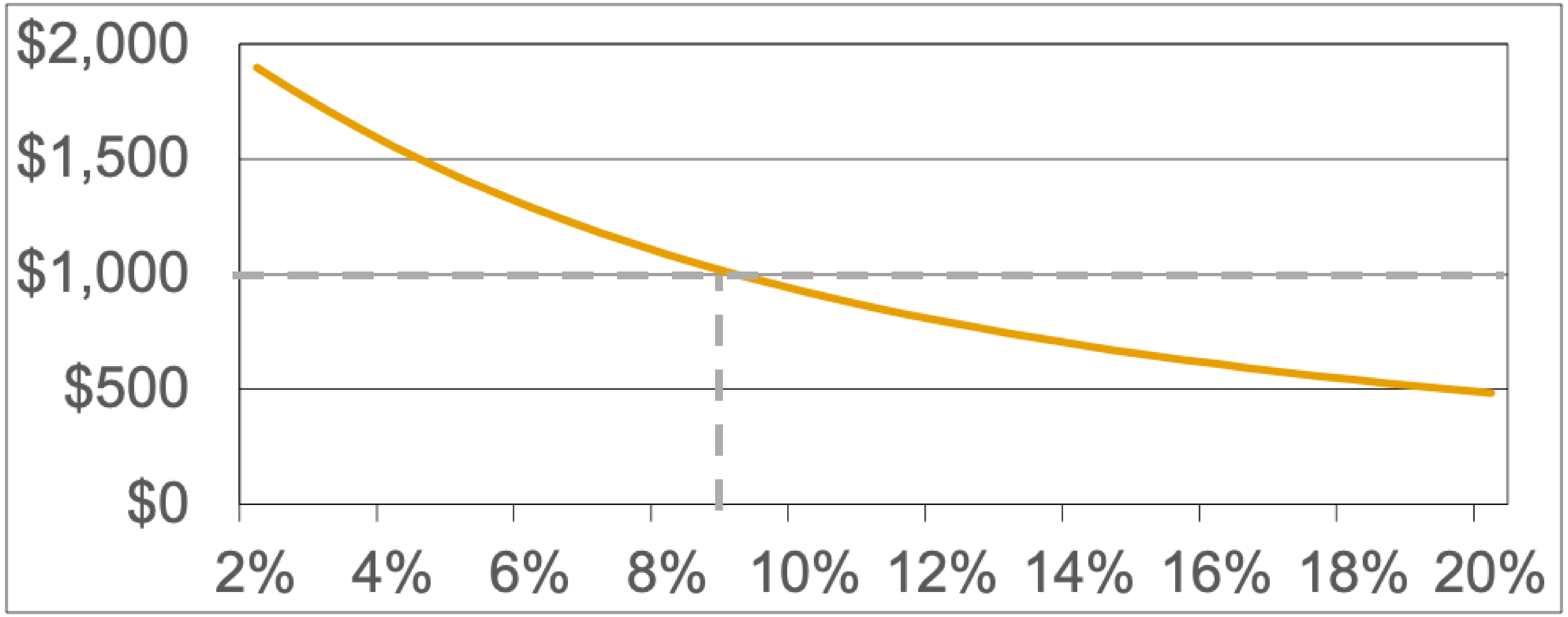
Bond Price (B)



Interest Rate (r)

A bond is selling at a…

Bond Price (B)



Interest Rate (r)

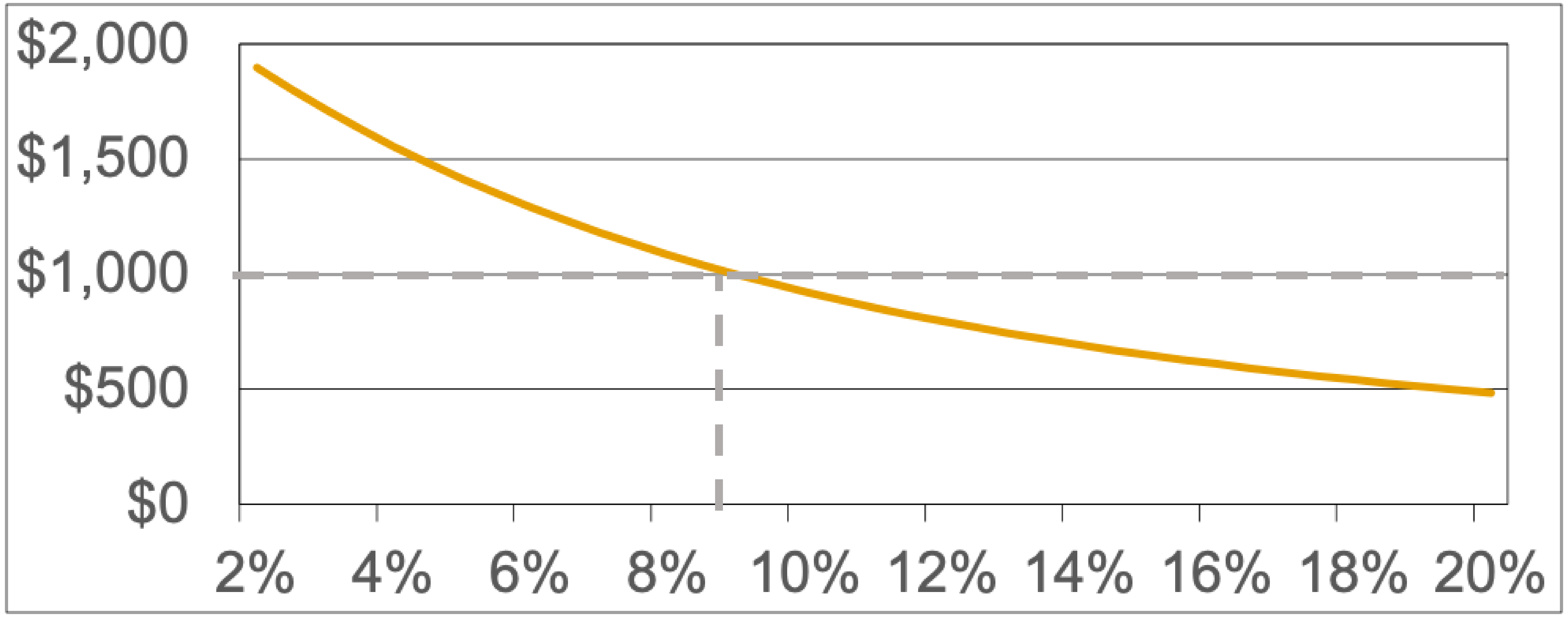
A bond is selling at a…

Premium
(B > F)

Discount
(B < F)

Bond Price (B)

Par Value
(B = F)



Interest Rate (r)

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Buying and Selling Bonds

Primary Market: Bonds issued for the first time (by country/corporate). Sold directly to investors.

Secondary Market: After bonds have been issued, are generally bought and sold between buyers and sellers ‘over-the-counter’ (OTC).

Secondary market bond prices are a percent of the bond’s par value.

For example, a bond trading at a market price of 102 would mean it is selling at 102% of its par value.

Bonds:
Yield to Maturity (YTM)

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Spot Rates

So far, we've assumed there's one interest rate that's constant over time.

In the real-world, this ***isn't*** the case.

The **'spot rate'** is the interest rate on a zero-coupon bond over a time horizon.

Use different spot rates to discount cash flows over different time horizons.

PV of bond paying $1 at end of one year:

PV of bond paying $1 at end of two years:

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Use different spot rates to discount cash flows over different time horizons.

PV of bond paying $1 at end of one year:

$$\boldsymbol{B = \frac{1}{(1+r_1)^1}}$$

PV of bond paying $1 at end of two years:

$$\boldsymbol{B = \frac{1}{(1+r_2)^2}}$$

Spot Rate -> Bond Price -> Yield to Maturity

PV of bond paying $1 at end of one year and $1 at end of two years:

If $r_1 = 3\%$ and $r_2 = 4\%$:

What is the “**yield**” if we held this bond to maturity (i.e., end of two years)?

Spot Rate -> Bond Price -> Yield to Maturity

PV of bond paying \$1 at end of one year and \$1 at end of two years:

$$\boldsymbol{B} = \frac{\mathbf{1}}{(\mathbf{1}+\boldsymbol{r}_{\mathbf{1}})^{\mathbf{1}}} + \frac{\mathbf{1}}{(\mathbf{1}+\boldsymbol{r}_{\mathbf{2}})^{\mathbf{2}}}$$

If $r_1 = 3\%$ and $r_2 = 4\%$:

$$\boldsymbol{B} = \frac{\mathbf{1}}{(\mathbf{1+0.03})^{\mathbf{1}}} + \frac{\mathbf{1}}{(\mathbf{1+0.04})^{\mathbf{2}}} = 1.895$$

What is the "**yield**" if we held this investment to maturity?

$$1.895 = \frac{\mathbf{1}}{(\mathbf{1}+\boldsymbol{YTM})^{\mathbf{1}}} + \frac{\mathbf{1}}{(\mathbf{1}+\boldsymbol{YTM})^{\mathbf{2}}} = 3.66\%$$

Yield to Maturity (YTM) is Internal Rate of Return (IRR) if you hold to maturity

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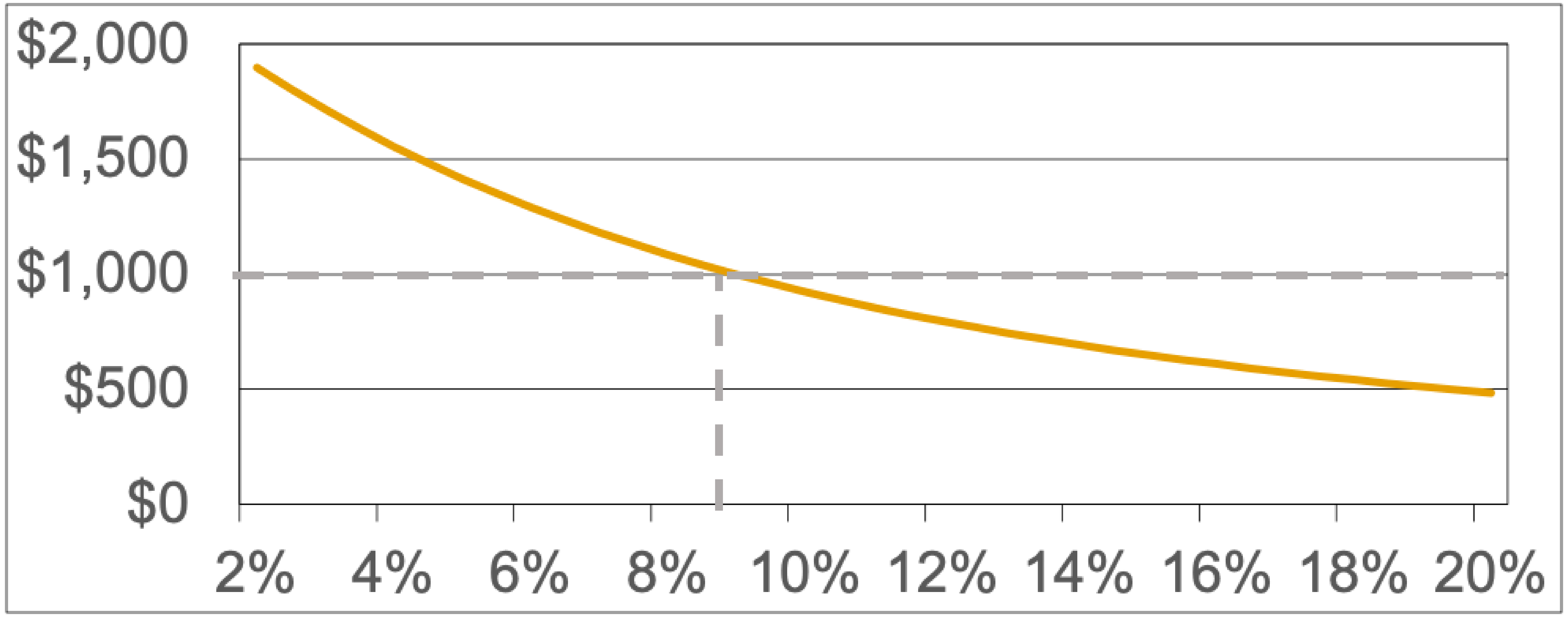
$$\boldsymbol{B = \frac{C}{YTM}\left(1 - \frac{1}{(1+YTM)^T}\right) + \frac{F}{(1+YTM)^T}}$$

Yield to Maturity (YTM) is Internal Rate of Return (IRR) if you hold to maturity

$$B = \frac{C}{YTM}\left(1 - \frac{1}{(1+YTM)^T}\right) + \frac{F}{(1+YTM)^T}$$

Bond Price (B)



Interest Rate (r)

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Yield to Maturity (YTM) is Internal Rate of Return (IRR) if you hold to maturity

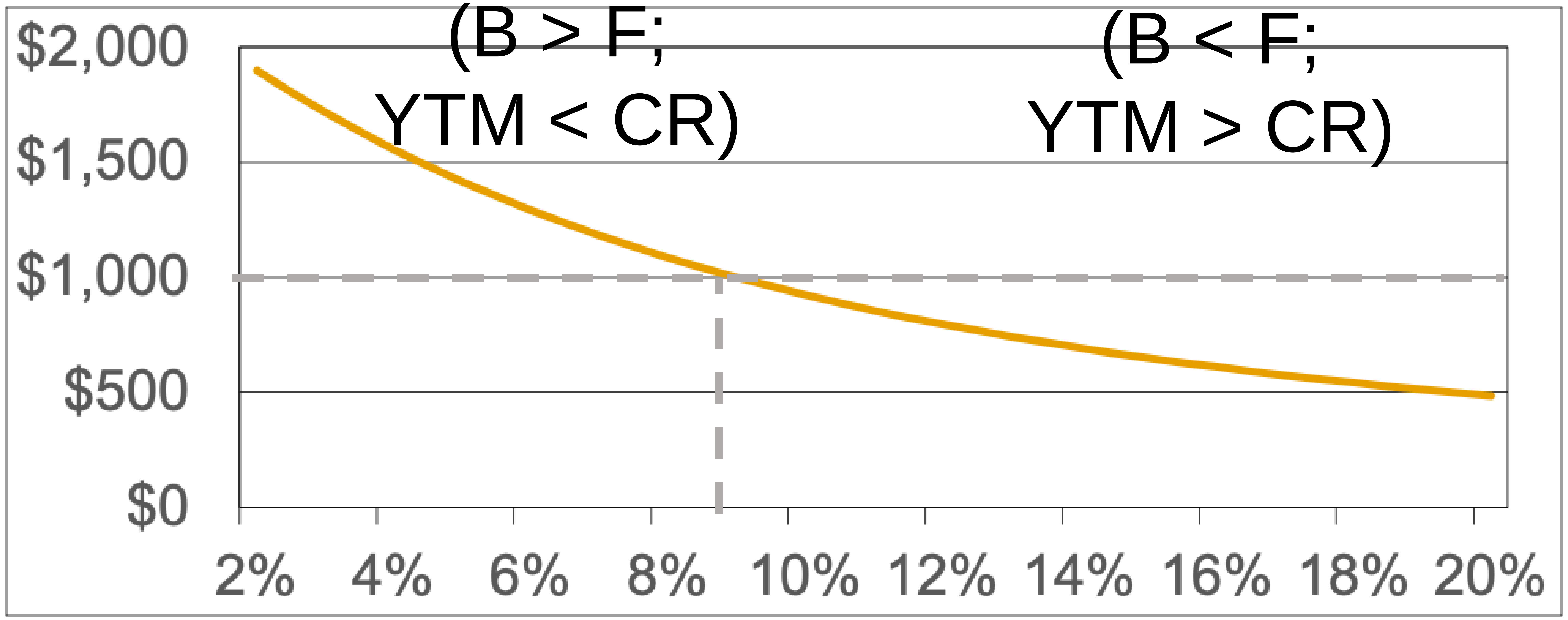
$$B = \frac{C}{YTM}\left(1 - \frac{1}{(1+YTM)^T}\right) + \frac{F}{(1+YTM)^T}$$

Premium (B > F; YTM < CR)

Discount (B < F; YTM > CR)

Bond Price (B)

Par Value (B = F; YTM = CR)



Interest Rate (r)

What is higher Yield to Maturity (YTM) compensating bond investors for?

What is higher Yield to Maturity (YTM) compensating bond investors for?

Time Value of Money + Risk Premia

- **Inflation**
- **Interest Rate Risk**
- **Default Risk**

Why might your return on a bond differ from the YTM?

Why might your return on a bond differ from the YTM?

If you sell it before maturity, the return may differ.

YTM assumes that any bond coupons are reinvested, but if they are put into different assets they may earn different returns.

If the bond defaults, you may get nothing or only a fraction of the principal.

US Treasuries

US Treasuries

Treasury Bills (T-Bills)

- Zero-coupon bonds. Range from 4 to 52 weeks to maturity. $100 increments

Treasury Notes (T-Notes)

- 2 to 10 years to maturity

Treasury Bonds (T-Bonds)

- 20 to 30 years to maturity

Treasury STRIPS (T-Strips)

- Long-term zero-coupon bonds created by separating coupons from T-notes/bonds

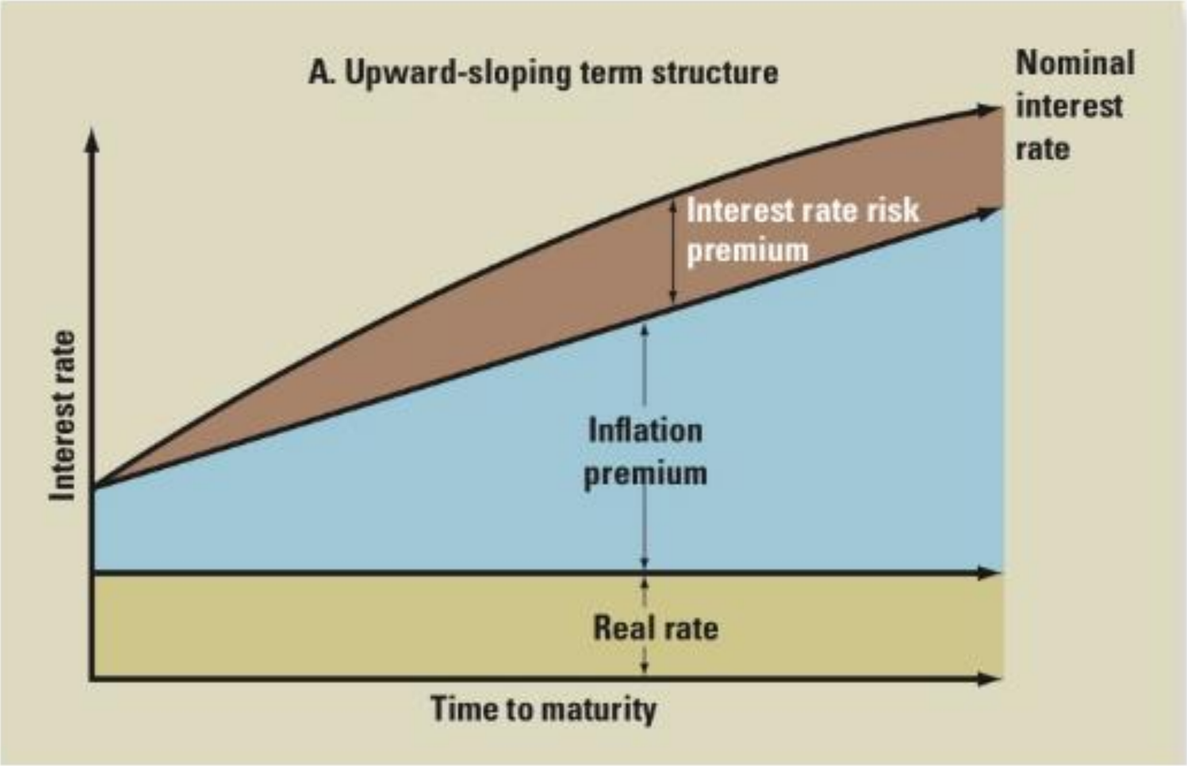
Treasury Inflation Indexed Securities (TIIS) / Inflation-Protected Securities (TIPS)

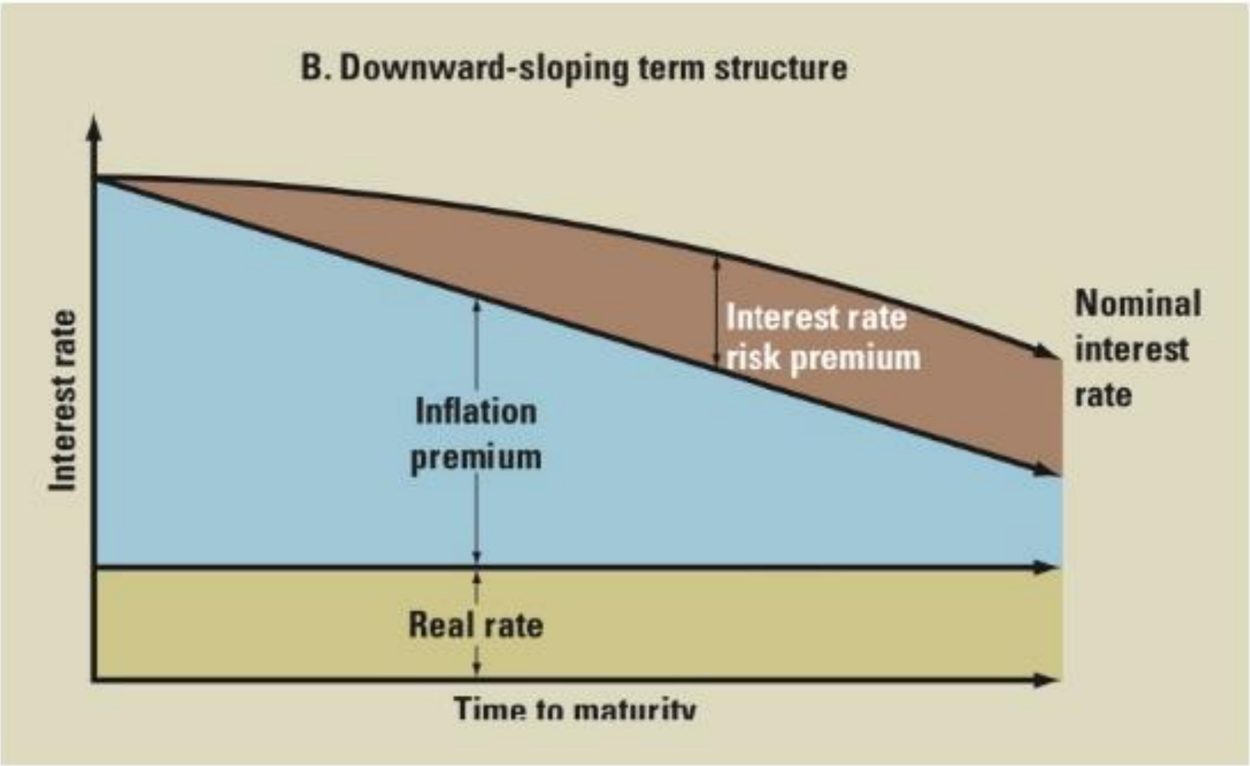
- Principal amount (F) adjusted for inflation, Coupons are %F so $C increase with inflation

Term Structure of Interest Rates / Yield Curve

- At a point-in-time, how do interest rates vary for bonds of different maturities?
- So far, we've assumed the interest rate is the same for all maturities. Flat "yield curve"
- Instead, different interest rates ("Spot Rates") for bonds of different maturities.
- Inverted yield curve is a recession indicator.

	Jan 15, 2024	May 1981	Average (1970-2013)
3 Month	5.45	16.45	6.36
6 Month	5.16	14.90	7.01
2 Year	4.14	13.87	7.51
3 Year	3.92	13.51	7.69
5 Year	3.84	13.09	7.91
10 Year	3.96	12.74	8.20
30 Year	4.20	12.49	8.34





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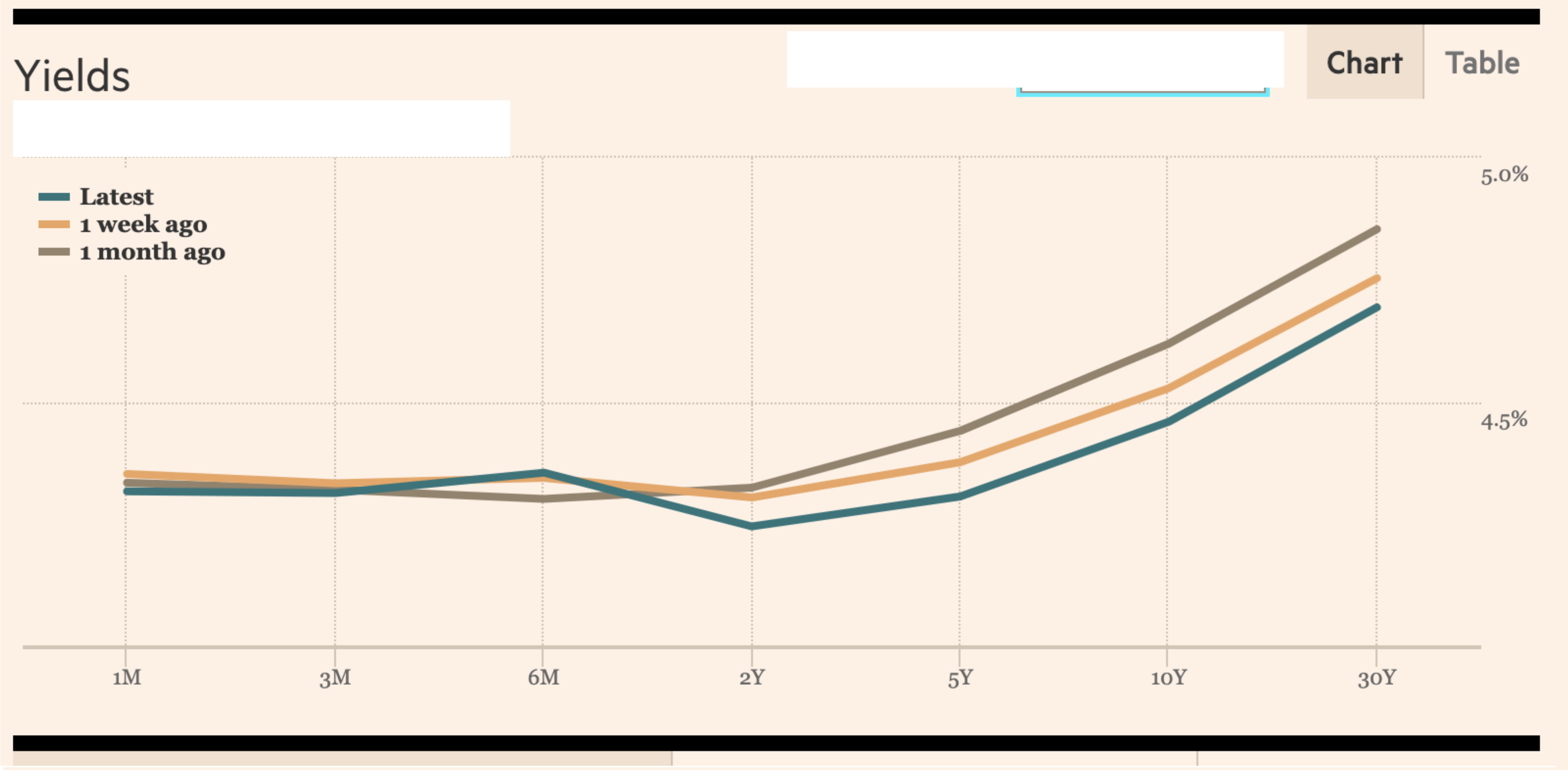
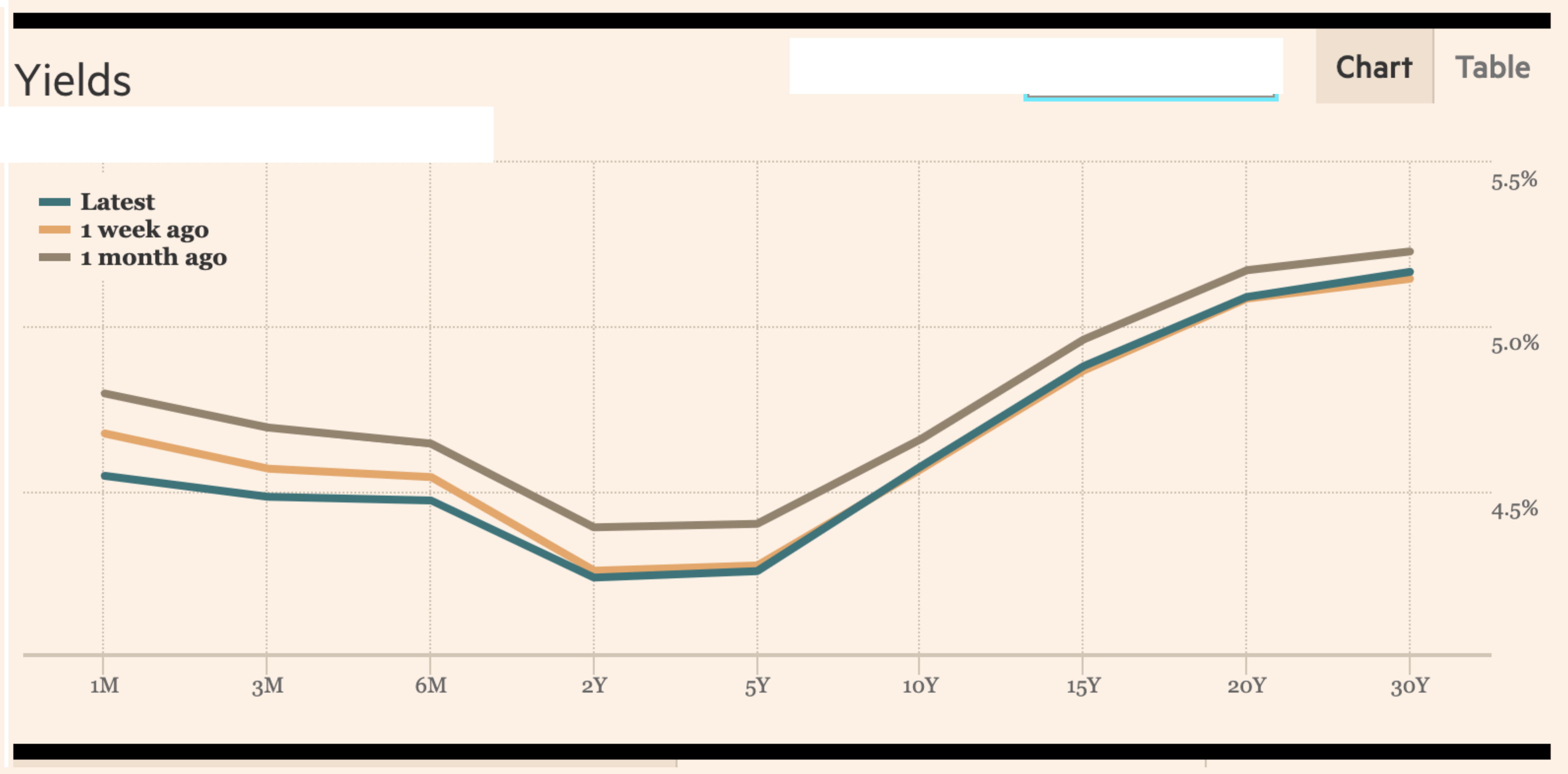
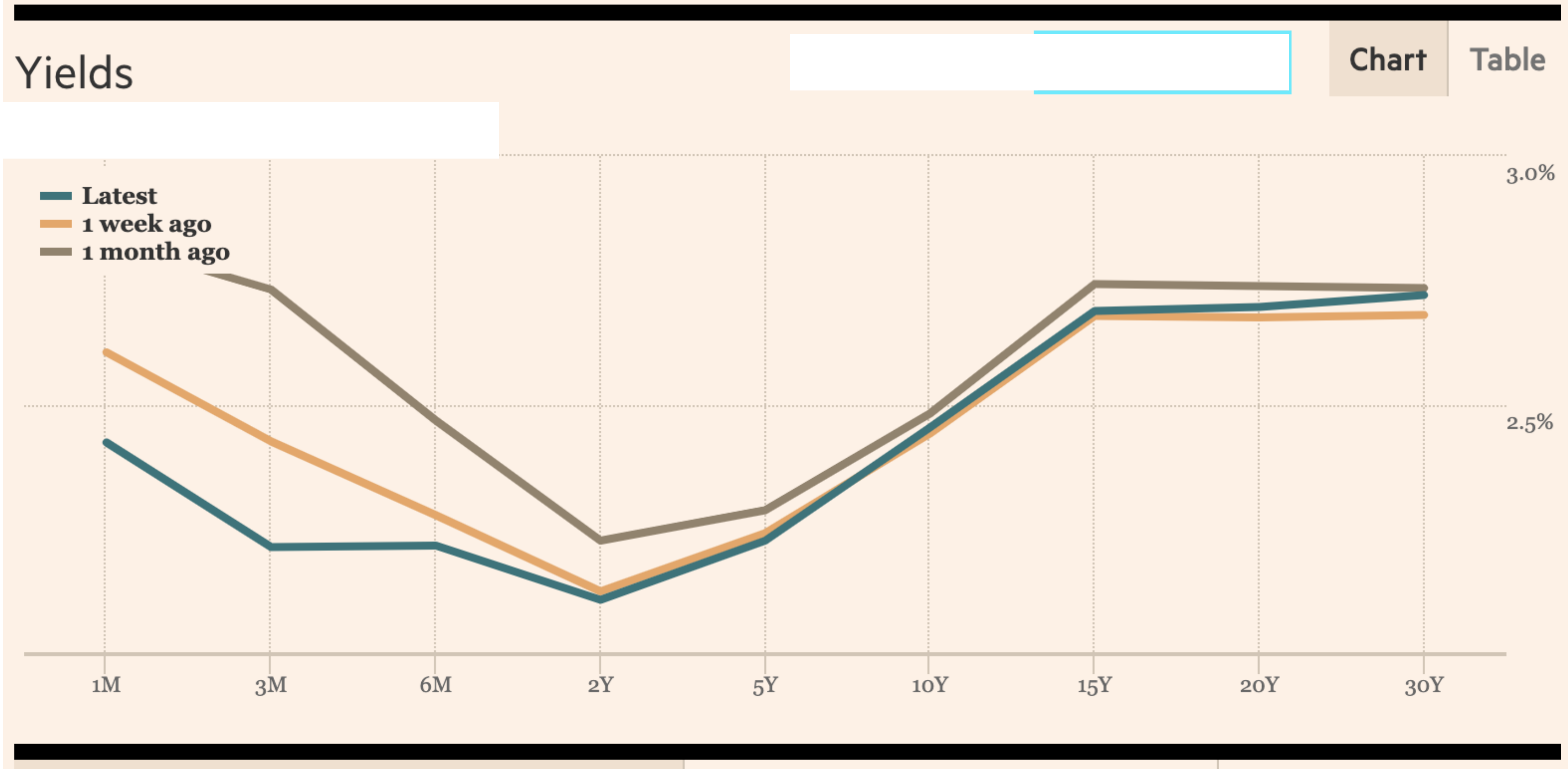
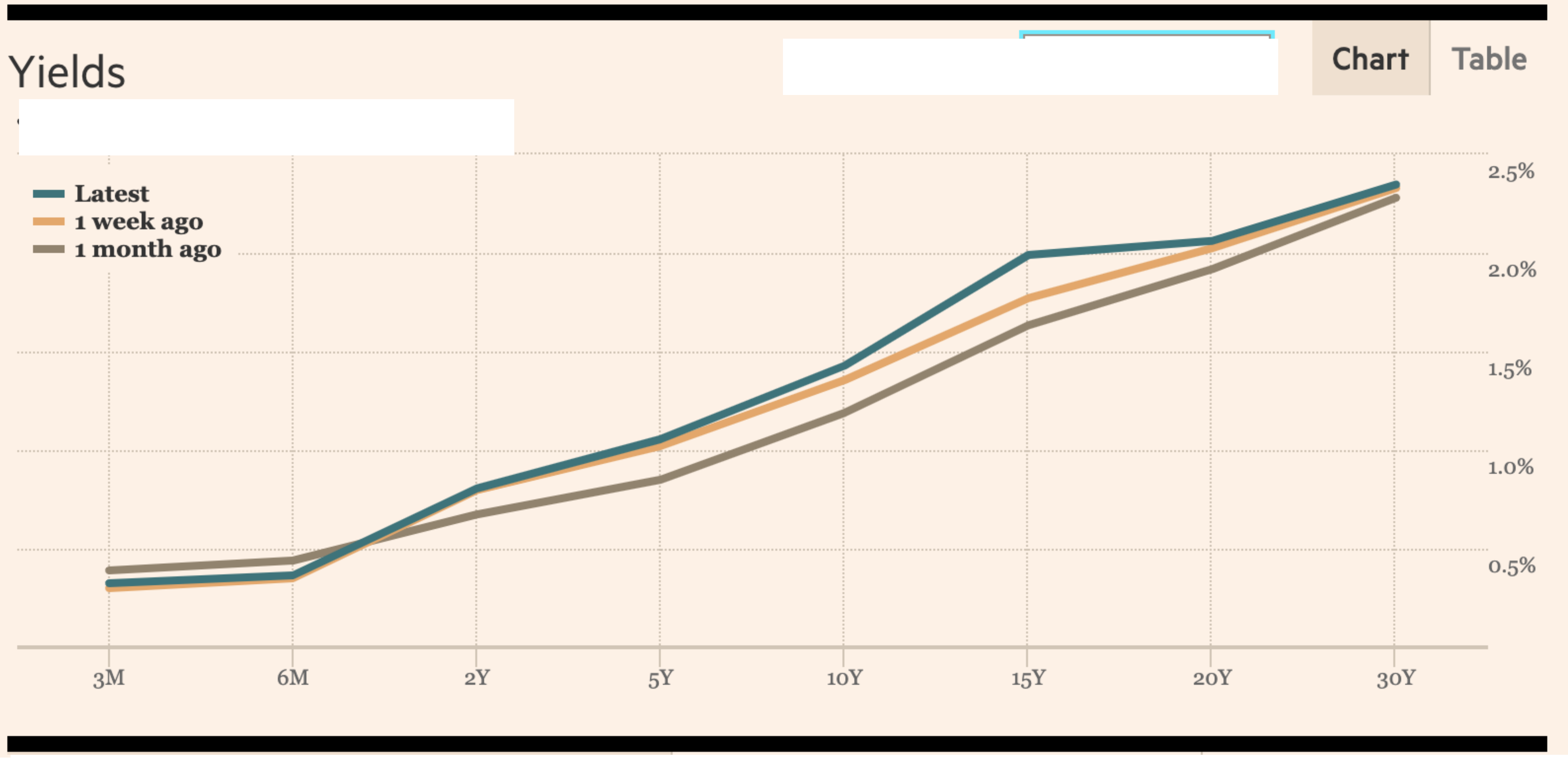
Yield Curves: What's happening now?

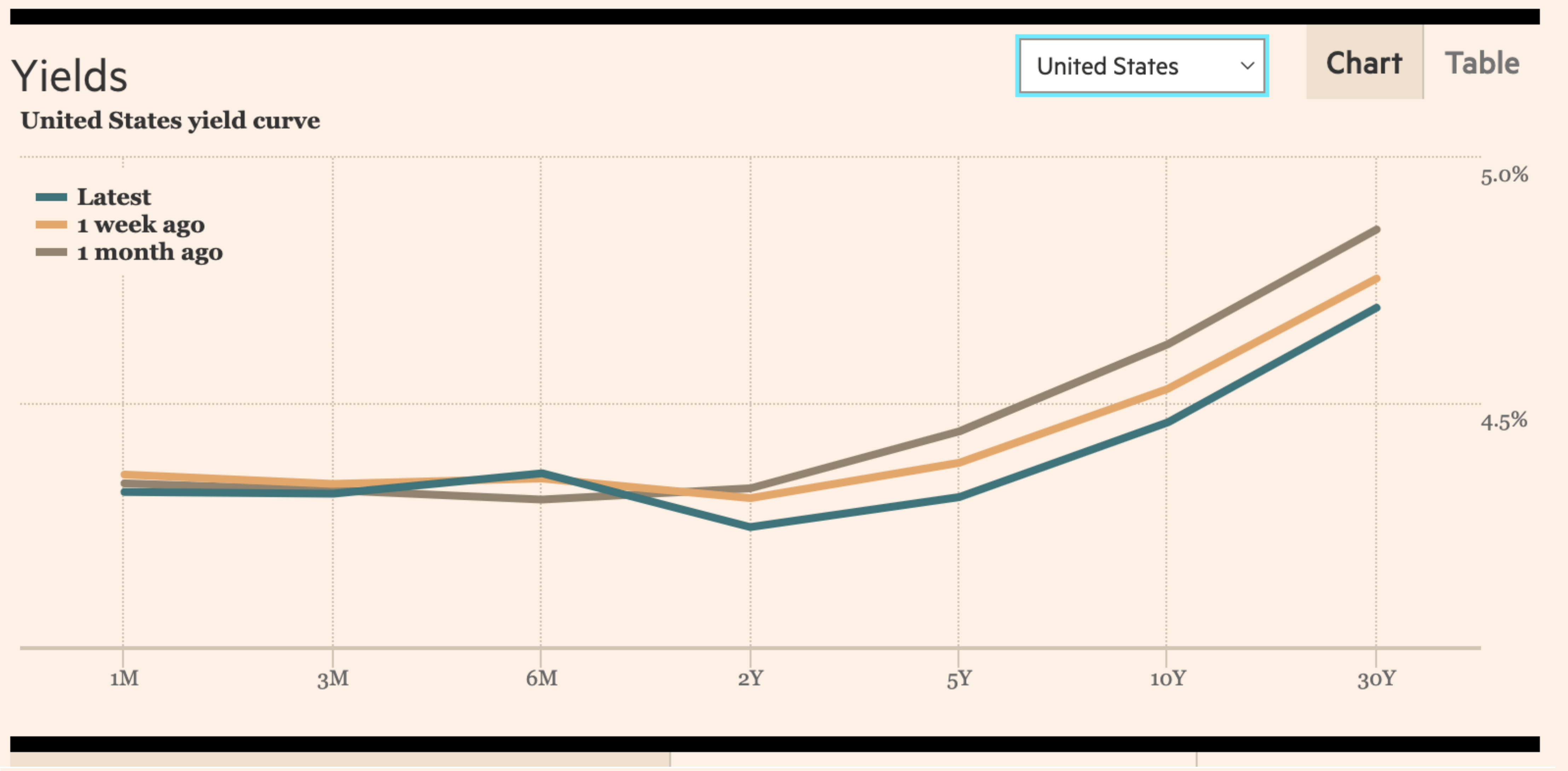


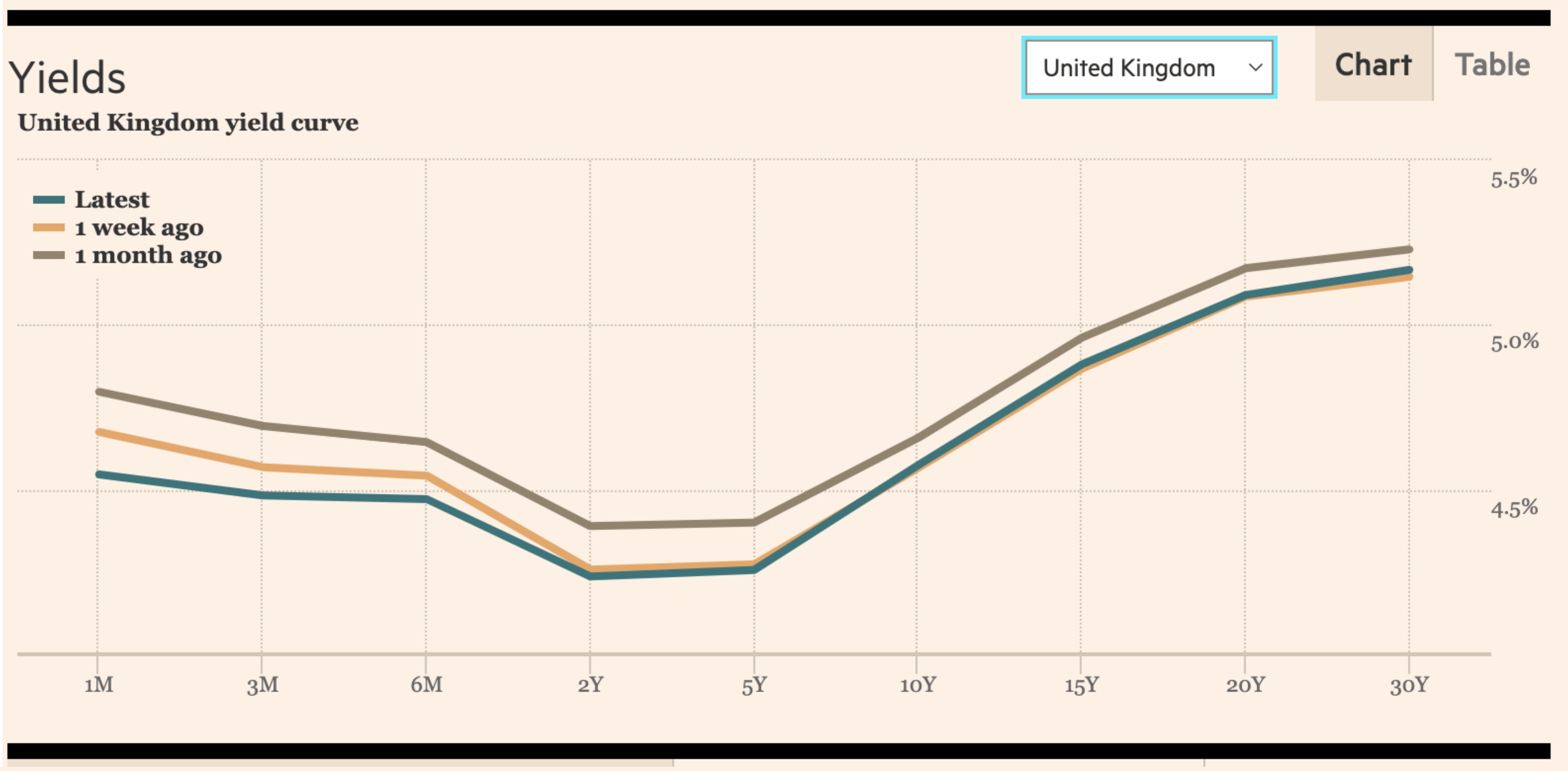


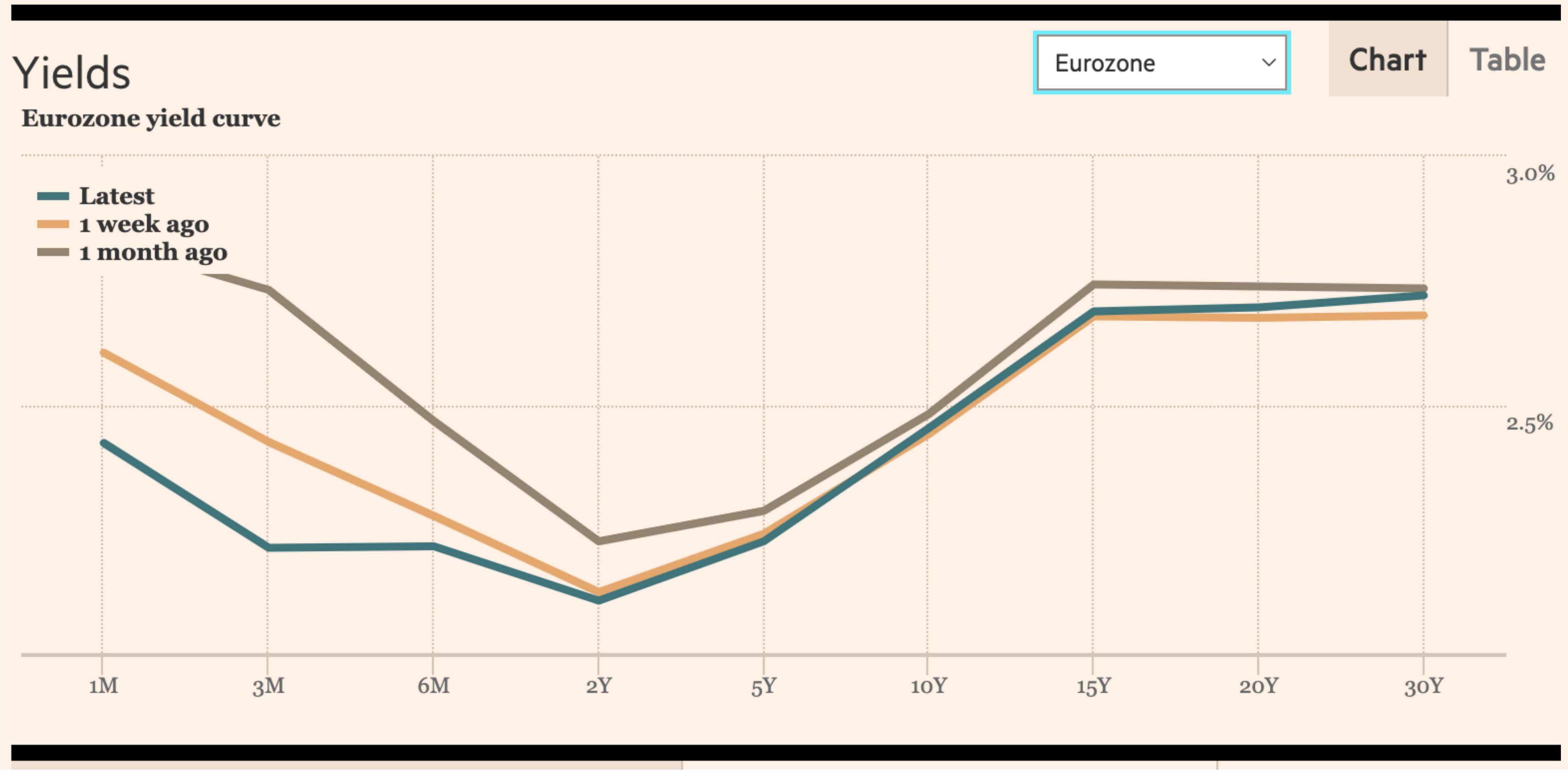


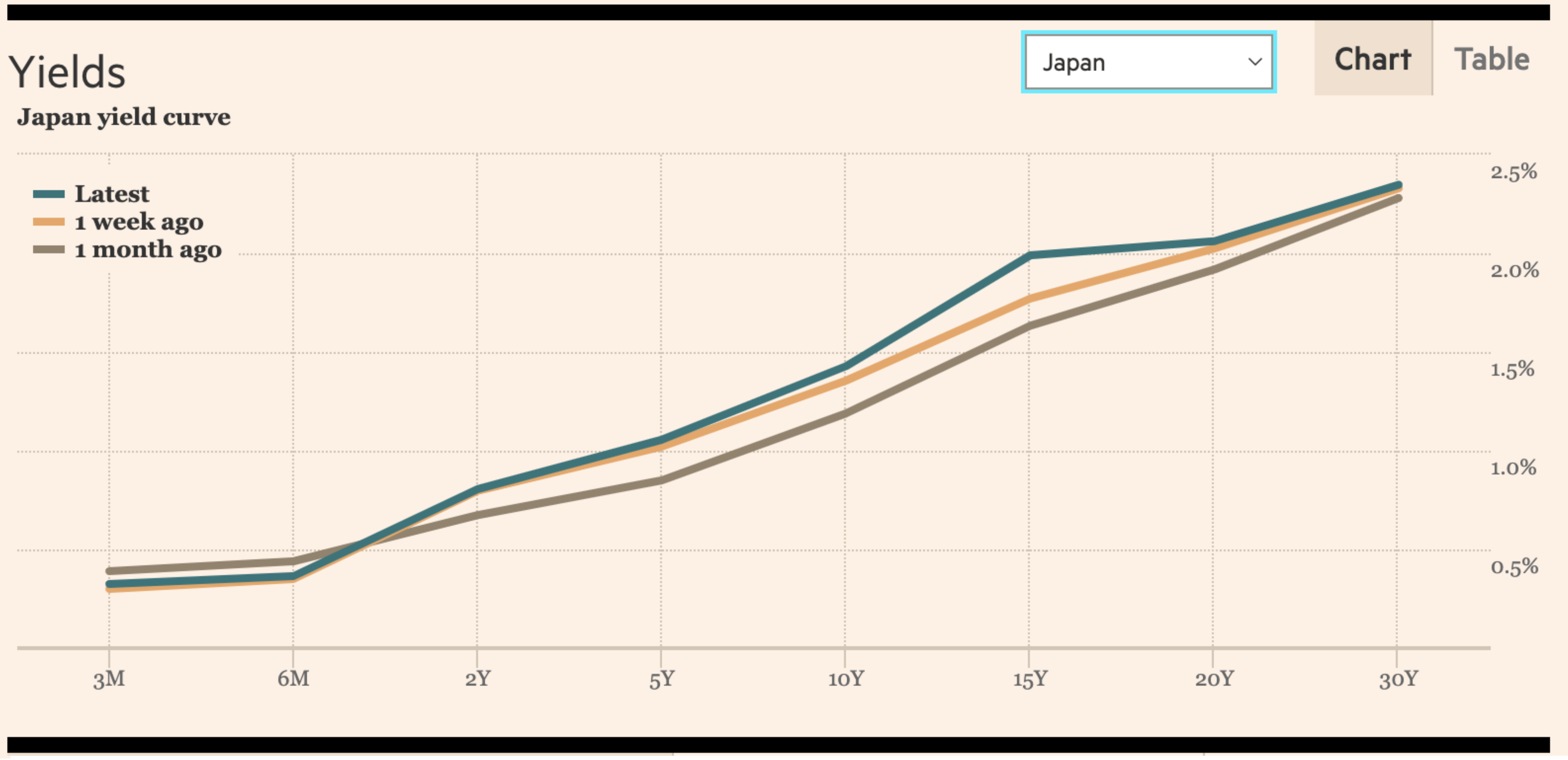


Yield Curves: What's happening now?





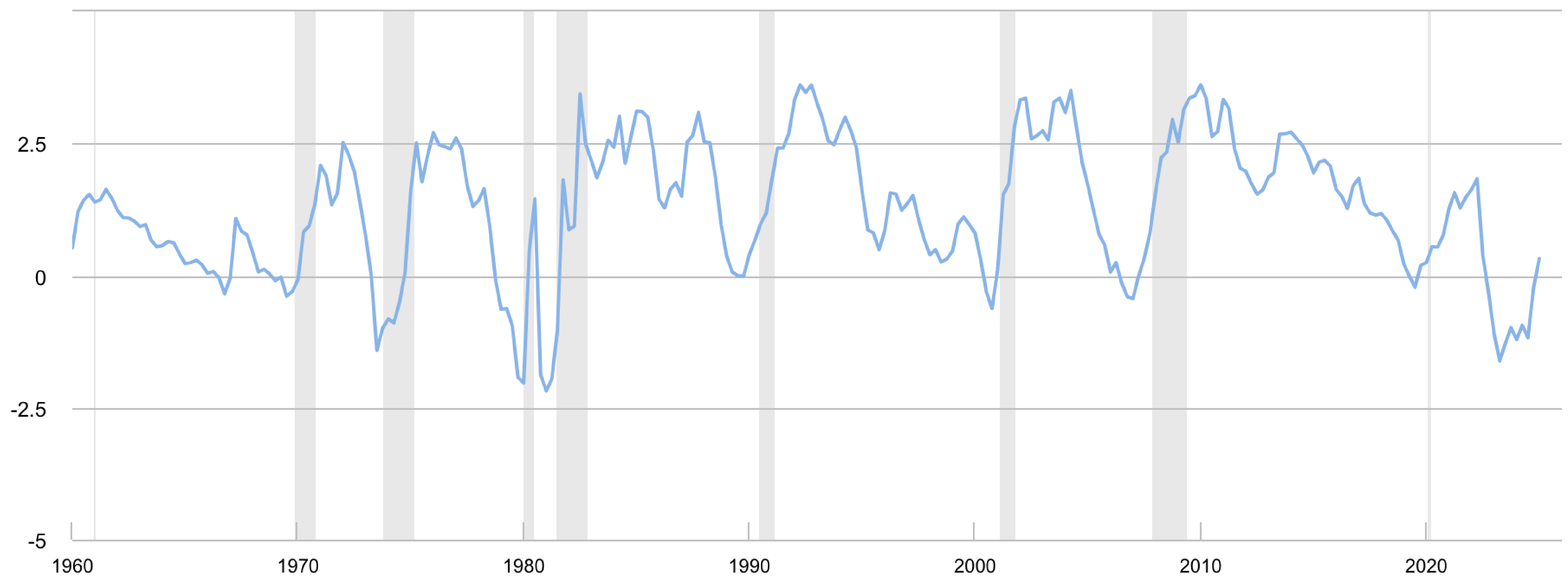




Slope of Yield Curves: Recession Indicator

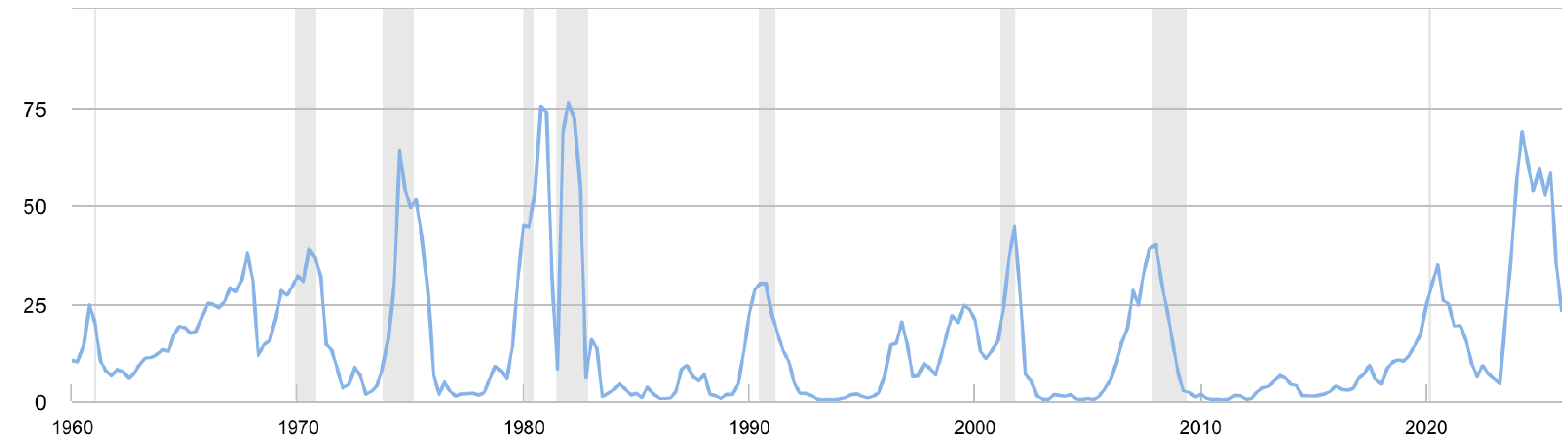
Treasury Term Spread: 10 Year Bond Rate - 3 Month Bill Rate

Percentage points (monthy average)



Probability of U.S. Recession, Twelve Months Ahead of Term Spread Readings

Percent (monthy average)



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Bond Payoff Structure and Ratings

Who Gets Paid? Bond Payoff Structure

Is it historically safer to hold a US treasury bond or a Venezuelan government bond?

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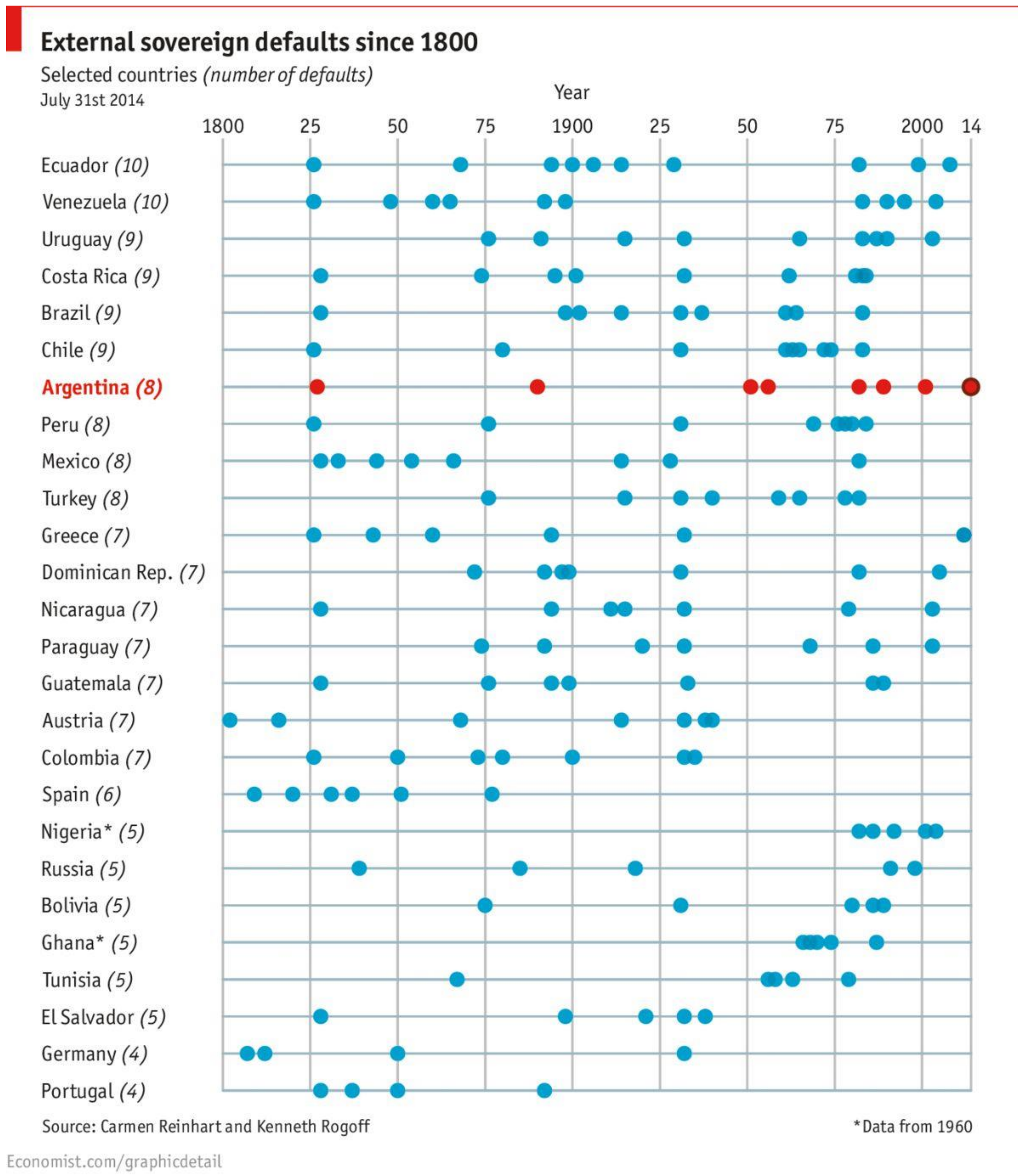

Bonds vary in their default risk

- Sovereign bonds issued by countries, municipal bonds by government entities, corporate bonds by firms.
- Different countries / firms vary in their risk of default.
- Riskier bonds have higher yields to compensate investors for higher default risk.
- Credit rating agencies (Moody's, Standard & Poor's, Fitch) assign **ratings**. Typically, higher rating -> lower yield.
- "Spreads" are relative yield over government bonds

Moody's	**Standard & Poor's / Fitch**	
Aaa	AAA	Investment Grade (IG)
Aa	AA	
A	A	
Baa	BBB	
Ba	BB	High Yield / Speculative / Junk
B	B	
Caa	CCC	
Ca	CC	
C	C	

Historic default rates on bonds

Global financial services average cumulative default rates (1981- 2022) (%)															
	--Time horizon (years)--														
Rating	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**	**Y8**	**Y9**	**Y10**	**Y11**	**Y12**	**Y13**	**Y14**	**Y15**
AAA	0.00	0.04	0.21	0.38	0.55	0.72	0.80	0.89	0.93	0.97	1.02	1.06	1.10	1.15	1.19
AA	0.03	0.09	0.17	0.30	0.44	0.57	0.67	0.78	0.88	0.97	1.04	1.10	1.15	1.21	1.26
A	0.07	0.18	0.31	0.44	0.57	0.71	0.86	0.97	1.06	1.17	1.26	1.33	1.40	1.47	1.56
BBB	0.21	0.55	0.92	1.29	1.60	1.88	2.12	2.35	2.61	2.82	3.07	3.33	3.60	3.90	4.19
BB	0.59	1.45	2.35	3.17	3.93	4.68	5.24	5.61	5.93	6.20	6.54	6.83	6.92	6.96	7.02
B	2.37	4.84	6.77	8.14	9.27	10.29	10.98	11.71	12.46	13.19	13.93	14.44	14.86	15.32	15.66
CCC/C	16.89	21.85	24.58	26.68	27.97	28.63	29.56	30.77	31.29	31.87	31.87	32.20	32.89	33.66	34.08
Investment grade	0.10	0.25	0.44	0.64	0.82	1.00	1.15	1.29	1.42	1.55	1.67	1.77	1.89	2.00	2.12
Speculative grade	2.17	3.90	5.31	6.43	7.37	8.21	8.85	9.41	9.91	10.37	10.84	11.21	11.46	11.71	11.89
All rated	0.48	0.92	1.32	1.68	1.99	2.27	2.51	2.71	2.90	3.07	3.24	3.38	3.51	3.64	3.77

Includes the bank, nonbank financial institution, and insurance sectors. Sources: S&P Global Ratings Credit Research & Insights and S&P Global Market Intelligence's CreditPro®.

Examples

1. You are consulting for a firm that is looking to build a factory. If they buy the land and build a factory, it will cost them $370,000. A local real estate agent projects that you can sell this land and building for $420,000 in one year. (Assume this is for certain). Suppose that the rate of interest is 5%. Should you invest?

a) Yes
b) No
c) Indifferent
d) Not enough information

2. A new project will need an investment of $2.4 million and will generate $1.2 million cash flows for 3 years. Calculate the NPV if the cost of capital is 15%.

3. Pinewood Studios is considering which new studio to invest in. Both studios will generate the same revenues. Which is a better investment at an 8% discount rate?
A) Studio Alec costs $150,000 and is expected to last 4 years with operating costs of $24,000 per year.
B) Studio Boris costs $100,000 and is expected to last 2 years with operating costs of $9,000 per year.

a) Yes
b) No
c) Indifferent
d) Not enough information

4. What is the market price of a zero coupon bond that matures in three years if the market's required return is 8%?

5. What is the price of a bond issued by Oddjob Corporation that pays $100 every year for 4 years, and also pays $1,000 at the end of four years? Assume the discount rate is 12%.

6. An investment has the following after-tax cash flow. What is the **Net Present Value (NPV)** of each project if the discount rate is 10%? Which would you select?

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV	IRR	PP	DPP	ROI
A	-200	209	0	0	0					
B	-200	0	140	140	140					
C	-200	70	70	70	140					
D	-200	0	0	0	200					
E	-200	0	0	0	400					
F	-400	0	0	0	800					

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7. An investment has the following after-tax cash flow. What is the **Internal Rate of Return (IRR)** of each project if the discount rate is 10%? Which would you select?

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV	IRR	PP	DPP	ROI
A	-200	209	0	0	0					
B	-200	0	140	140	140					
C	-200	70	70	70	140					
D	-200	0	0	0	200					
E	-200	0	0	0	400					
F	-400	0	0	0	800					

8. An investment has the following after-tax cash flow. What is the **Payback Period (PP)** of each project if the discount rate is 10%? Which would you select?

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV	IRR	PP	DPP	ROI
A	-200	209	0	0	0					
B	-200	0	140	140	140					
C	-200	70	70	70	140					
D	-200	0	0	0	200					
E	-200	0	0	0	400					
F	-400	0	0	0	800					

9. An investment has the following after-tax cash flow. What is the **Discounted Payback Period (DPP)** of each project if the discount rate is 10%? Which would you select?

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV	IRR	PP	DPP	ROI
A	-200	209	0	0	0					
B	-200	0	140	140	140					
C	-200	70	70	70	140					
D	-200	0	0	0	200					
E	-200	0	0	0	400					
F	-400	0	0	0	800					

10. An investment has the following after-tax cash flow. What is the **Return on Investment (ROI)** of each project if the discount rate is 10%? Which would you select?

Project	CF_0	CF_1	CF_2	CF_3	CF_4	NPV	IRR	PP	DPP	ROI
A	-200	209	0	0	0					
B	-200	0	140	140	140					
C	-200	70	70	70	140					
D	-200	0	0	0	200					
E	-200	0	0	0	400					
F	-400	0	0	0	800					

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